

The learned suitability surface: how it is built and evaluated

Rwanda malaria exposure model — pixel surface $S(\text{temp}, \text{NDVI}, \text{amp})$

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What we are trying to build

We want a **suitability surface** S : a function that takes an environmental state — here a triple of (*temperature, annual-mean NDVI, NDVI seasonal amplitude*) — and returns a number we interpret as **malaria risk** for living under that state. Once we have S , two things follow:

1. a **pixel risk map** — evaluate S at every 250 m pixel's environment; and
2. a **sector prediction** — combine the surface with where people live to get a predicted disease level per administrative sector, which we can check against observed incidence.

The difficulty: **we never observe risk at the pixel level**. We only observe **disease counts aggregated to sectors** (≈ 404 of them). So we cannot fit S pixel-by-pixel. The trick described below recovers a pixel-level S *purely from sector-level outcomes*, by exploiting one structural assumption that makes the problem linear.

The ingredients

symbol	meaning	source
p	a pixel (250 m)	MODIS / CHLSA grid
T_p, N_p, A_p	temperature, NDVI, NDVI amplitude at pixel p	<code>stack.nc</code> (2021)
w_p	population in pixel p	WorldPop 2021
$\text{sec}(p)$	the sector pixel p belongs to	rasterized boundaries
y_s	observed malaria incidence (cases/1000) in sector s	DHIS2
$d(s)$	the district sector s belongs to	DHIS2 parent

NDVI amplitude is $A_p = \max_{\text{month}} \text{NDVI} - \min_{\text{month}} \text{NDVI}$ over 2021 — a measure of how much the vegetation dries down seasonally (the moisture-stress axis).

The one assumption that makes it work

We assume a sector's risk is the **population-weighted average of a per-environment suitability**. In words: a sector is risky to the extent that *its people live in risky environments*. Formally, if pixel p has suitability $S(T_p, N_p, A_p)$, the sector's predicted risk is

$$\hat{y}_s = \frac{\sum_{p \in s} w_p S(T_p, N_p, A_p)}{\sum_{p \in s} w_p}. \quad (1)$$

This is the *entire* model. Everything else is bookkeeping to turn (1) into something we can fit.

The key observation: **equation (1) is linear in the unknown surface S** . S appears only multiplied by known weights and summed. Linear models are easy to fit. We just have to write (1) in matrix form.

Step 1 — Discretise the environment into cells

We cannot estimate S at every possible real-valued triple, so we **bin** each axis and assume S is constant within a cell. With the equal-width bins of the 3D surface:

- temperature: 8 bins, edges 10, 12, ..., 26 °C
- NDVI: 6 bins, edges 0.00, 0.15, ..., 0.90
- amplitude: 5 bins, edges 0.00, 0.12, ..., 0.60

That is a grid of $8 \times 6 \times 5 = 240$ **cells**. Write $c(p)$ for the cell pixel p falls into, and let S_c be the (unknown) suitability of cell c . The surface is now just a **vector of 240 numbers** $\mathbf{S} = (S_1, \dots, S_{240})$.

Step 2 — Summarise each sector by a population histogram

For each sector s we count **how much population sits in each cell**:

$$H_s(c) = \sum_{\substack{p \in s \\ c(p)=c}} w_p, \quad p_s(c) = \frac{H_s(c)}{\sum_{c'} H_s(c')}. \quad (2)$$

p_s is the **population distribution of sector s over the 240 cells** — the fraction of the sector's people living in each environment. By construction $\sum_c p_s(c) = 1$. In code this is one `np.bincount` over the flattened cell index, then a normalisation (`Hn` in the script).

Each sector is now a single row of 240 numbers. Stack all sectors: $\mathbf{X} \in \mathbb{R}^{404 \times 240}$, row s equal to p_s .

Step 3 — The model is a matrix product

Substitute the binned surface into (1). Because S is constant within a cell, the population-weighted average becomes a weighted sum over cells:

$$\hat{y}_s = \sum_c S_c p_s(c) = \langle \mathbf{p}_s, \mathbf{S} \rangle, \quad \text{i.e.} \quad \hat{\mathbf{y}} = \mathbf{X} \mathbf{S}. \quad (3)$$

So the predicted-incidence vector is **just $\mathbf{X}$ times the surface**. This is an ordinary linear regression with the unknown surface $\mathbf{S}$ playing the role of the regression coefficients, and the per-sector histograms playing the role of the features.

This is the crux. We did not fit a pixel surface and then average it. We defined each sector by *where its population sits in environment-space*, and asked linear regression: *what risk value must each environment cell carry so that the population-weighted cell mix predicts each sector’s incidence?* The answer **is** the surface.

A 2-cell worked example

Suppose only two cells, “cool” and “warm”. Two sectors:

- Sector A: 10% of people cool, 90% warm $\Rightarrow \mathbf{p}_A = (0.1, 0.9)$
- Sector B: 70% cool, 30% warm $\Rightarrow \mathbf{p}_B = (0.7, 0.3)$

Then $\hat{y}_A = 0.1 S_{\text{cool}} + 0.9 S_{\text{warm}}$ and $\hat{y}_B = 0.7 S_{\text{cool}} + 0.3 S_{\text{warm}}$. Given the observed incidences y_A, y_B , regression solves these for $(S_{\text{cool}}, S_{\text{warm}})$ — the two cells’ risk. With 404 sectors and 240 cells it is the same thing at scale.

Step 4 — Fit with ridge regression (and why)

We have 240 unknowns and ~404 sectors, but the columns of $\mathbf{X}$ are highly **collinear** (neighbouring cells co-occur in the same sectors) and many cells are nearly **empty** (almost nobody lives at 12 °C or NDVI 0.85). Plain least squares would be unstable — it would fit wild, oscillating risk values to empty cells.

We therefore use **ridge regression**, which penalises large surface values:

$$\hat{\mathbf{S}} = \arg \min_{\mathbf{S}} \underbrace{\sum_s (y_s - \langle \mathbf{p}_s, \mathbf{S} \rangle)^2}_{\text{fit}} + \alpha \underbrace{\sum_c S_c^2}_{\text{smoothness / shrinkage}}. \quad (4)$$

The penalty weight α trades fit against stability. Large α shrinks the whole surface toward a flat constant (under-fit); small α lets it chase the data (risk of over-fit). We report a small sweep $\alpha \in \{1, 5, 20, 50, 100\}$ rather than fixing one value, so the comparison between surfaces is not an artefact of a lucky penalty.

(Note: because every row of $\mathbf{X}$ sums to 1, adding a constant to all S_c shifts every prediction equally — that constant is absorbed by the regression intercept, and the risk **ranking** we ultimately score is unaffected by it.)

Step 5 — Evaluate honestly: leave-one-district-out

If we fit $\mathbf{S}$ on all sectors and then scored it on those same sectors, we would be grading the model on data it has seen. Instead we use **leave-one-district-out (LODO)** cross-validation. The ~ 404 sectors group into 46 districts. For each district d :

1. fit the surface $\hat{\mathbf{S}}^{(-d)}$ using only sectors **not** in d (equation 4 on the training sectors);
2. predict the held-out district’s sectors: $\hat{y}_s = \langle \mathbf{p}_s, \hat{\mathbf{S}}^{(-d)} \rangle$ for $s \in d$.

Collect the out-of-fold predictions for all sectors and compute one number:

$$\rho = \text{Spearman}(\hat{\mathbf{y}}_{\text{oof}}, \mathbf{y}).$$

Grouping by **district** (not random sectors) is deliberate: it forces the model to generalise to a *spatially new region*, the honest “would this work somewhere we haven’t calibrated” test.

We score with **Spearman rank correlation** (not R^2) because we care about getting the **ordering** of sectors right — which are higher-burden than which — and Spearman is invariant to any monotone distortion of the scale and robust to the handful of sectors with implausible incidence (facility-catchment denominators).

Step 6 — Read out the surface and the map

Two distinct objects come out, and it is worth keeping them straight:

- **The skill numbers** (ρ) come from the *out-of-fold* predictions (Step 5). This is what we trust as “how good is the model”.
- **The displayed surface and pixel map** are fit *once on all sectors* (good for visualisation, not a skill claim). The surface is $\hat{\mathbf{S}}$ reshaped back to the $8 \times 6 \times 5$ grid; the pixel map assigns each pixel its cell’s value, $\text{map}(p) = \hat{S}_{c(p)}$.

Results

Leave-one-district-out Spearman vs incidence, equal-width bins, ridge α sweep (best in **bold**):

surface	$\alpha=1$	5	20	50	100	best
1D	0.577	0.571	0.541	0.484	0.459	0.577
$S(\text{temp})$						
2D	0.696	0.688	0.673	0.652	0.627	0.696
$S(\text{temp}, \text{NDVI})$						
3D	0.758	0.742	0.709	0.677	0.628	0.758
$S(\text{temp}, \text{NDVI}, \text{amp})$						

The 1D/2D rows are the same machinery with the histogram **marginalised** over the unused axes (sum p_s over NDVI and/or amplitude before fitting). Adding the seasonal-amplitude axis lifts the surface from 0.696 to **0.758** — matching the full sector random-forest model (~ 0.75) but using only two satellite raster stacks and remaining fully interpretable.

Binning choice changes the answer

The bin edges are themselves a modelling choice. Equal-width bins waste resolution on near-empty environments; **equal-population (quantile) bins** put a bin boundary where people are, so each bin holds the same number of people:

surface	equal-width (best)	equal-population (best)
1D $S(\text{temp})$	0.577	0.646
2D $S(\text{temp}, \text{NDVI})$	0.696	0.714
3D $S(\text{temp}, \text{NDVI}, \text{amp})$	0.758	0.752

Equal-population binning sharply helps the low-dimensional surfaces (the 1D gain is almost entirely better bin *placement*, not new signal) but does nothing for the 3D surface and makes its ridge fit more sensitive to α — denser, more collinear histograms destabilise under heavy shrinkage.

Things to keep in mind

- **The surface only “sees” inhabited cells.** $\sim 83\%$ of people live at 18–24 °C and $\sim 74\%$ in NDVI 0.45–0.60, so most of the 240 cells are empty and carry no fitted weight (the pale regions of the surface plot). The amplitude axis is the most *population-dispersed* of the three, which is why it adds discriminating power.
- **In-sample surface $\neq$ out-of-fold skill.** The picture is fit on everything; the ρ values are cross-validated. Do not read a skill claim off the map.
- **Linearity is an assumption.** Equation (1) says a sector’s risk is a *linear* population-weighted mean of cell suitabilities. The surface itself can be arbitrarily nonlinear in (temp, NDVI, amp) — that nonlinearity is captured by letting each cell have its own value — but interactions *between sectors* or saturation effects are not modelled.
- **Single year (2021).** The whole vegetation/amplitude analysis uses one year; confirming the amplitude gain across multiple years is the main outstanding robustness check.
- **A data seam.** The MODIS NDVI mosaic has an ~ 8 -pixel all-year gap (a tile seam) that shows as a thin NaN line on the maps; it is excluded from every sector histogram and does not affect the statistics.